

TABLE II
THICKNESS RANGE

PI Foil thickness	Thickness range	Cost	Lead Times
Usual PI thickness	25µm/50µm	Standard	Standard
Intermediate PI thickness	12µm/75µm/100µm	+	+
Unusual PI thickness	9µm/125µm/150µm	2 to 3 times the cost of usual thickness	2 to 4 months

Unsymmetrical structure. Buildup is not symmetrical in z axis.

Designing rigid-flex PCBs

Raw material. Polyimide must be used for flex parts. Standard FR4 or high Tg material can be mixed with polyimide layers for rigid parts. Fully-polyimide material can be also considered for high temperature.

Prepreg is recommended to be no-flow around flex area.

TABLE III
PROPERTIES OF COPPER

Copper property	Electro-deposited				Rolled anodised		
	1/4oz (9µM)	1/2oz (18µM)	1oz (35µM)	2oz (70µM)	1/2oz (18µM)	1oz (35µM)	2oz (70µM)
Tensile strength (KPSI)	15	33	40	40	20	22	28
Elongation (%)	2	2	3	3	8	13	27
Vol resistivity (MOHM CM)	—	1.66	1.62	1.62	1.78	1.74	1.74

TABLE IV
TYPICAL DESIGN GUIDELINES

Guideline	Good	Bad
Conductors on flex areas should be staggered, alternating their locations.		
To increase copper strength, design pads larger than access holes.		
Acute angles at the interface between conductors and pads concentrate stress. Avoid this using fillets. All pads—through-hole and SMT—should be filleted. Pad fillets improve etched yield and material strength.		
Having a radiused corner within flex bend area reduces or eliminates stress concentrators and improves reliability.		
Apply crosshatching to help the circuit retain flexibility.		
Incorporate tear stops and reliefs for slits in the circuit to prevent tear from starting and propagating.		
Copper strands should be perpendicular to bending lines, to prevent tearing and scratching.		

Uniflow prepreg is possible on other layers.

Rigid-flex PCB. Rigid-flex PCB must use adhesive-less foils. Adhesive-less foils are 30 to 50 per cent more expensive than adhesive foils. However, it is mandatory for the plating process. Using adhesive-less foils is expressly recommended by IP 2223 C.

Rigid PCB. As for rigid-flex PCBs, two kinds of copper layers exist. One, rolled anodised copper, which is more ductile and better for dynamic flexion. Second, electro-deposited copper that has better resistance and is better for fine lines.

Rolled anodised copper is manufactured by lamination between rolls. Whereas, electro-deposited copper is manufactured by copper electro-deposition.

Basic design rules. There can be many possible designs and stackups of rigid-flex PCBs, just as there can be many possible errors while designing these. Material is costly and, hence, design must help save costs by achieving the highest possible yield. PI material provides the best thermal protection, but it has the worst mechanical properties.

Given below are basic guidelines for designing rigid-flex PCBs.

Baking before soldering. The main issue is moisture absorption. Complete baking is required before soldering.

Flexibility and bending radius.

- Single side: Six times flex thickness
- Double side: 12 times flex thickness
- Multi-layered: 24 times flex thickness

Flexible solder mask or PI coverlay.

- Flexible solder mask for no dynamic flexion
- PI coverlay for dynamic flexion

Conclusion

Even if rigid-flex PCBs are more expensive than rigid boards, the solution is a global one. Assembly cost can be lower than when using a large number of PCB types. Labour cost for manual operation can be avoided. And, reliability can be increased. That is a real gain. Rigid-flex PCBs are more compact and technically advanced. **EFY**